

(b) Suggest a procedure for a general proof.

Solution: Let $F(a_1, a_2, \dots, a_n)$ be the given Boolean function. List down the 2^n minterms in ascending order. Form 2^{n-1} consecutive pairs of minterms and by algebraic manipulation, remove the literal a_n . Again form 2^{n-2} consecutive pairs of minterms and by algebraic manipulation, remove the literal a_{n-1} . Repeat this process until the expression $a_1 a_1'$ remains. By postulate 5(a), $a_1 a_1' = 1$. $\square$

2-17: The product of all minterms of a Boolean function of n variables is 0.

(a) Prove the above statement for $n=3$.

Solution: Let $F(A, B, C)$ be the Boolean function. The minterms are: $(A+B+C)$, $(A+B+C')$, $(A+B'+C)$, $(A+B'+C')$, $(A+B+C)$, $(A+B+C')$, $(A+B'+C)$, $(A+B'+C')$.

$$(A+B+C) \cdot (A+B+C') = (A+B+C \cdot C') \text{ (Postulate 4(b))}$$

$$= (A+B+0) \text{ (Postulate 5(b))}$$

$$= (A+B) \text{ (Postulate 2(a))}$$

$$(A+B'+C) \cdot (A+B'+C') = (A+B'+C \cdot C') \text{ (Postulate 4(b))}$$

$$= (A+B'+0) \text{ (Postulate 5(b))}$$

$$= (A+B') \text{ (Postulate 2(a))}$$

$$(A+B+C) \cdot (A+B'+C) = (A+B+C \cdot C') \text{ (Postulate 4(b))}$$

$$= (A+B+0) \text{ (Postulate 5(b))}$$

$$= (A+B) \text{ (Postulate 2(a))}$$

$$(A+B+C) \cdot (A+B'+C') = (A+B+C \cdot C') \text{ (Postulate 4(b))}$$

$$= (A+B+0) \text{ (Postulate 5(b))}$$

$$= (A+B) \text{ (Postulate 2(a))}$$

$$(A+B) \cdot (A+B') = (A+B \cdot B') \text{ (Postulate 4(b))}$$

$$= (A+0) \text{ (Postulate 5(b))}$$

$$= A \text{ (Postulate 2(a))}$$